Catherine Hae Seung Jeon

Google Scholar — ORCID — GitHub — LinkedIn — Website

EDUCATION

❖ **University of Chicago** — *Illinois, United States*
Ph.D. in Computer Science (Advisor: Prof. Ben Y. Zhao) — *09/2026 — Present*
- Fellowship: Liew Family Fellowship

❖ **Ewha Womans University** — *Seoul, South Korea*
M.S. in Computer Science & Engineering (Advisor: Prof. S.E. Oh) — *09/2024 — 08/2026*
- GPA: 4.26/4.30

❖ **Ewha Womans University** — *Seoul, South Korea*
B.S. in Computer Science & Engineering — *03/2020 — 08/2024*
- GPA: 4.15/4.30
- Honors: Dean's List (7 semesters), Summa Cum Laude

PUBLICATIONS

❖ Peer-Reviewed Conference Publications

Haeseung Jeon, Yeonseo Cho, Nate Mathews, Rob Jansen, Marc Juarez, and Se Eun Oh (2026). "Brewing Under Pressure: More Realistic Tor Flow Correlation Considering Low FPRs". In *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security*. [CORE A*, Acceptance Rate=20%]

Jimin Park, **Haeseung Jeon**, AHyun Ji, Aritran Piplai, Mohammad Saidur Rahman, and Se Eun Oh (June 9, 2026). "TabCL: Continual Malware Classification with Tabular-Aware Generation". In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pp. 547–559. [CORE B, Acceptance Rate=25%]

Haeseung Jeon*, Sujin Kim*, Nate Mathews, Hosung Kang, and Se Eun Oh (Apr. 12, 2026). "RoFiRe: Robust Website Fingerprinting on Real-World Tor Traffic via Improved Augmentation and Normalization". In *Proceedings of the ACM Web Conference 2026*, pp. 3335–3346. [CORE A*, Acceptance Rate=20%]

Chaiwon Hwang*, **Haeseung Jeon***, Jiwoo Hong, Hosung Kang, Nate Mathews, Goun Kim, and Se Eun Oh (Aug. 24, 2025). "Enhancing Search Privacy on Tor: Advanced Deep Keyword Fingerprinting Attacks and BurstGuard Defense". In *Proceedings of the 20th ACM Asia Conference on Computer and Communications Security*, pp. 621–634. [CORE A, Acceptance Rate=20%]

❖ Peer-Reviewed Journal Publications

Noori Kim, **Catherine H. S. Jeon**†, and Mingsian R Bai (Apr. 28, 2026). "Impedance-Based Acoustic Load Inference Using Intelligent Speaker Sensors". In *IEEE Sensors Journal*. [SCIE, Q1, IF=4.5 (JCR 2025)]

Noori Kim, Keisuke Alexander Nakamura, Max Chen, **Catherine H. S. Jeon**†, and Mingsian R Bai (Mar. 23, 2026). "Speakers as sensors: Machine learning-based earphone impedance analysis for ear canal insertion depth prediction". In *The Journal of the Acoustical Society of America* 159.3, pp. 2729–2740. [SCIE, Q1, IF=2.6 (JCR 2025)]

❖ Short Paper/Poster Publications

Haeseung Jeon, Jiwoo Hong, Saeyeon Hong, Hosung Kang, Bona Kim, Se Eun Oh, and Noori Kim (Aug. 22, 2025). "Domain-adapted automatic speech recognition with deep neural networks for enhanced speech intelligibility prediction". In *Proceedings of the 6th Clarity Workshop on Improving Speech-in-Noise for Hearing Devices*, pp. 15–17. [Short Paper] [Challenge Rank Top 5]

Chaiwon Hwang*, **Haeseung Jeon***, Goun Kim, Jiwoo Hong, Hosung Kang, and Se Eun Oh (Dec. 12, 2024b). "Securing Search Privacy on Tor: Deep Keyword Fingerprinting and BurstGuard Defense". In *2024 Annual Computer Security Applications Conference*. [Poster]

Chaiwon Hwang*, **Haeseung Jeon***, Goun Kim, Jiwoo Hong, Hosung Kang, and Se Eun Oh (June 26, 2024a). "DKF: Employing Deep Learning for Keyword Fingerprinting Attacks on Tor".

*Equally contributing authors, listed in alphabetical order.
†Legal name in South Korea: Haeseung Jeon; United States: Catherine Hae Seung Jeon.

In *Proceedings of Korea Computer Congress 2024*, pp. 1381–1383. [Short Paper] [Distinguished Paper Award]

RESEARCH EXPERIENCE

❖ **UChicago | Security, Algorithms, Networking and Data Lab** *Illinois, United States*
Graduate Research Assistant (Advisor: Prof. Ben Zhao) *09/2026 — Present*

- Participating in research on AI agents' vulnerability analysis capabilities. Analyze reasoning and static analysis techniques within trajectories and compare them with traditional static analysis tool to explore agent's capabilities beyond human expertise.

❖ **Ewha Womans University | AI Security Lab** *Seoul, South Korea*
Graduate Research Assistant & Lab Head (Advisor: Prof. S.E. Oh) *09/2024 — 08/2026*

- Led the development of a few-shot learning model for website fingerprinting attack. Proposed a robust framework (pre-norm, RMSNorm, and adaptive window augmentation), delivering up to 9% gains over a SimCLR-based baseline and achieving new SOTA.
- Drove development of a feature embedding model for flow correlation attack. Designed a CvT-style architecture with online mining and KDE-based feature representation. Achieved 88% TPR at an extremely low FPR, establishing the real-world feasibility of the attack.

Undergraduate Research Assistant *01/2023 — 08/2024*

- Led research on the first DL-based website fingerprinting model, achieving 41-55% gains over a previous SOTA. Built a Selenium crawler to collect 300K+ fresh samples over 1+ years, profiled unique data patterns, and benchmarked ML/DL baselines on a new dataset.

❖ **Purdue University | Sensor, Electro-Acoustics Technology Lab** *Indiana, United States*
Visiting Researcher (Research Advisor: Prof. N. Kim) *01/2026 — 05/2026*

- Designed a CNN-based physical-basis (e.g., acoustic load) prediction model using the electroacoustic coupling characteristic of earphones. Led customer discovery of the proposed solution through NSF Regional I-Corps, resulting in acceptance into the National I-Corps.

Part-Time Researcher *05/2025 — 01/2026*

- Led a team in the 3rd Clarity Prediction Challenge (CPC3), developing a speech intelligibility prediction model for hearing-aid evaluation. Designed and implemented a fundamental architecture (ASR-backboned Transformer) and training pipeline, achieving 5th place.

❖ **UTEP | Intelligent & Quantum Secure Advanced Cyber Defense Lab** *Texas, United States*
Visiting Researcher (Research Advisor: Prof. M.S. Rahman) *06/2025 — 08/2025*

- Built a continual learning framework with a conditional tabular GAN and improved storage/training efficiency by generative replay. Drove overall experiments and achieved up to 16% improvements over the baselines (GAN and TVAE) in Class-IL/Time-IL scenarios.

AWARDS & SCHOLARSHIPS

UChicago, Liew Family Fellowship — *USD $8K (5 yrs.)* — *FA 2026 — FA 2030*
EWU, Outstanding Academic Paper Award — *KRW ₩1M (≈ USD $750)* — *SP 2026*
Boeing Korea, Ewha-Boeing Scholarship — *KRW ₩4M (≈ USD $3K)* — *SP 2026*
EWU, Outstanding Ewha Scientist Scholarship — *half-tuition (2 yrs.)* — *FA 2024 — SP 2026*
Korea Computer Congress, Distinguished Paper Award — *SP 2024*
EWU, Software Future Talent Scholarship — *KRW ₩1M (≈ USD $750)* — *SP 2024*
Huawei Korea, Huawei ICT Talent Scholarship — *KRW ₩1M (≈ USD $750)* — *FA 2023*
EWU, Honors Scholarship (top 6%) — *quarter-tuition* — *FA 2023*
EWU, Highest Honors Scholarship (top 2%) — *half-tuition* — *FA 2020, SP 2022, SP 2023*
EWU, W4 Admissions Scholarship (top entrant) — *full-tuition (4 yrs.)* — *SP 2020 — SP 2024*

COMMUNITY ENGAGEMENT

EWU CS Undergraduate Lab Fair, Invited Speaker — *09/2025*
UTEP CS Summer Research Symposium, Invited Speaker — *08/2025*
EWU CS Undergraduate Open Lab Seminar, Invited Speaker — *03/2025*
GDSC Devfest π 2024: Responsible AI in action, Invited Speaker — *03/2024*
AWS Cloud Club (ACC), Cloud Team Committee Member — *09/2023 — 08/2024*
EWU Undergraduate Peer Tutoring Program, Lecturer — *06/2022, 06/2024*
Google Developer Student Clubs (GDSC), Backend Team Leader — *09/2022 — 07/2024*